

MINOR PROJECT REPORT

Implementing Basic Cloud Security Controls in Microsoft Azure

Design, Hardening, Identity & Storage Controls Security Assessment

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Executive Summary

In cloud-based environments, default configurations are frequently scanned and targeted by automated malicious entities. A fundamental security best practice is to deploy workloads with restricted public exposures, ensure strict segregation of administrative duties, and enforce access controls on corporate assets. This report documents the successful deployment, configuration, and hardening of a secure cloud infrastructure in Microsoft Azure for XYZ Solutions. This minor project establishes baseline preventive security controls including host port hardening, Least Privilege access permissions, and secure time-bound delegated storage access.

Section A: Theory Assessment

1. Explain the Shared Responsibility Model in cloud computing.

The Shared Responsibility Model splits security obligations between the Cloud Service Provider (CSP) and the customer. The CSP is responsible for the 'Security of the Cloud' (physical datacenters, virtualization infrastructure, host hardware, and physical networks). The customer is responsible for the 'Security in the Cloud'.

customer responsibility changes depending on the cloud service model:

- **Infrastructure as a Service (IaaS):** The customer maintains maximum control, including configuring operating systems, network firewalls (NSGs), middleware, databases, application code, identities, and raw data.
- **Platform as a Service (PaaS):** The CSP manages OS patching, hardware hosting, and database engines. The customer is responsible for configuring access policies, database structures, application code, and data objects.
- **Software as a Service (SaaS):** The CSP manages almost the entire stack. The customer remains responsible for data governance, endpoint devices, and access management (MFA, passwords).

2. What is the Principle of Least Privilege (PoLP) and why is it important?

The Principle of Least Privilege (PoLP) is a cybersecurity concept which states that users, automated service accounts, and active processes must only be granted the minimum set of permissions necessary to perform their assigned work functions.

Why PoLP is important because it:

- **Limits the Blast Radius:** If user credentials are stolen, the attacker's capability is bounded by the specific roles assigned to that identity, preventing tenant-wide takeover.
- **Reduces Human Error:** Restricts users from making accidental configuration errors, such as dropping database tables or deleting production virtual machines.

3. Differentiate between Public Blob Storage and Private Blob Storage.

Public Blob Storage and Private Blob Storage are two different configurations in Azure Storage that dictate how data is accessed over the internet:

- **Public Blob Storage:** Enables anonymous internet read access to files (blobs) or directories. Anyone with the URL can view or download the file without passing authentication checks. It is useful for public web assets, but presents high exposure risks.
- **Private Blob Storage:** Restricts all access to authorized users. Direct HTTP requests without active tokens return a 404 (Not Found) or 403 (Forbidden) response. Access requires Shared Access Signatures(SAS)

4. What is the purpose of a SAS Token in Azure Storage?

A Shared Access Signature (SAS) token is a query parameter appended to an Azure Storage URL that grants delegated, time-bound, and policy-restricted access to specific storage objects.

A SAS token enforces strict constraints, including allowed actions (e.g. read-only), specific expiration date and time, permitted IP ranges, and required protocol (enforcing HTTPS).

5. What is the difference between Azure Monitor and Microsoft Sentinel?

The primary difference lies in their operational focus and targeting scope:

- **Azure Monitor:** Focuses on operational health, diagnostic logging, and performance metrics (e.g., CPU, memory, uptime, latency). Its primary purpose is to ensure the reliability and speed of cloud workloads.
- **Microsoft Sentinel:** Focuses on cybersecurity and threat detection. It analyzes data ingested from Azure Monitor and other external logs to identify security anomalies, flag hacking patterns, manage alerts, and execute security playbooks.

Section B: Practical Implementation

System Security Architecture Diagram

The cloud environment was structured inside an isolated Resource Group named 'interns-rg'. The diagram below illustrates the boundaries, network security group rules, identity roles, and storage container permission boundaries deployed:

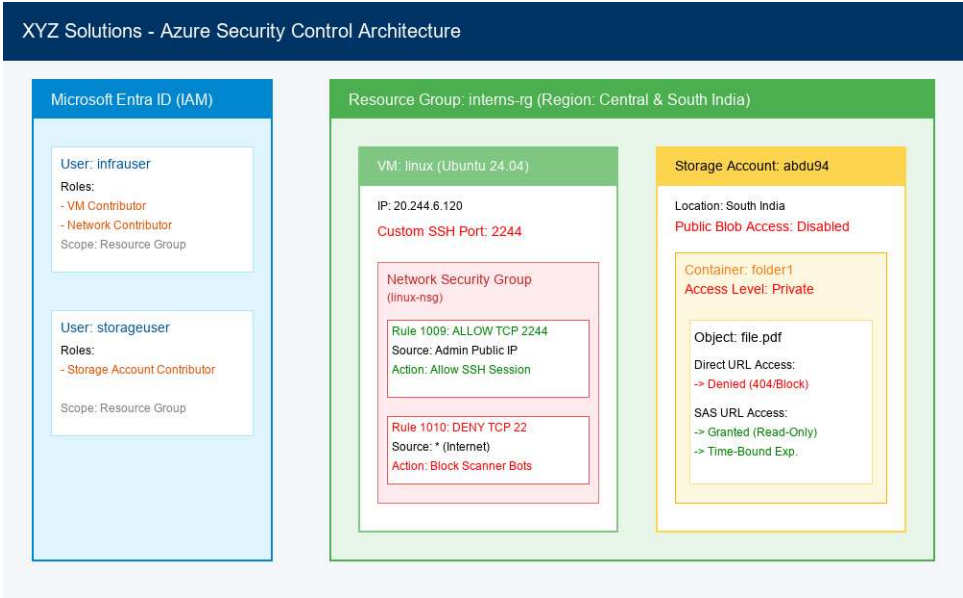


Figure A: System Architecture Diagram showing VM, network, IAM, and storage security perimeters.

Task 1: Secure Storage Account & SAS Access Configuration

A private container was created inside an Azure Storage Account to secure corporate files. Public anonymous access was disabled. Granular, time-bound access was configured using Shared Access Signatures (SAS).

Parameter	Configuration Detail
Storage Account Name	abdu94
Container Name	folder1
Access Level	Private (No anonymous public read access)
SAS Permission	Read-only
Signing Key	Key 1

Step-by-step Execution and Evidence:

1. Storage Account was created. A blob container named 'folder1' was configured with public access set to Private.

Home > Storage center | Blob Storage

Create a storage account

Basics Advanced Networking Data protection Security Enc

[View automation template](#)

Basics

Subscription	Azure subscription 1
Resource group	interns-rg
Location	South India
Storage account name	abdu94
Primary service	
Performance	Standard
Replication	Read-access geo-redundant storage (RA-GRS)

Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled

[Previous](#) [Next](#) [Create](#)

Figure 1: Essentials dashboard of Storage Account 'abdu94'.

abdu94 | Containers

Storage account

Search

[Add container](#) [Upload](#) [Refresh](#) [Delete](#) [Change access level](#)

Search containers by prefix

Showing all 1 items

	Name	Last modified	Anonymous access level
☐	\$logs	6/27/2026, 3:32:35 PM	Private

New container

Name *

folder

Anonymous access level

Private (no anonymous access)

Advanced

[Create](#) [Give feedback](#)

Figure 2: Creating container 'folder1' and checking its Private access properties.

2. A file named 'file.pdf' was uploaded to the container.

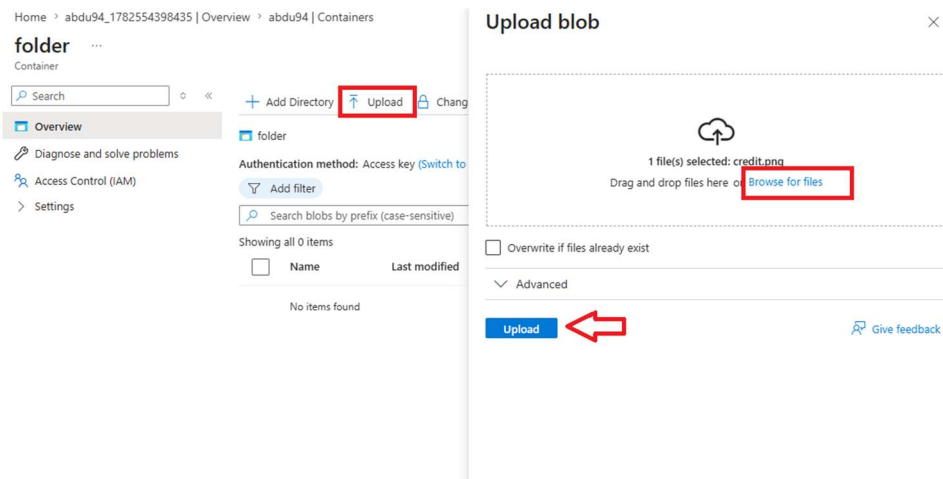


Figure 3: Container blob interface listing the uploaded file.

3. An anonymous read attempt was performed using the direct public URL. Access was denied, returning an XML error.



Figure 4: Public URL access attempt blocked, returning a ResourceNotFound/PublicAccessNotAllowed error.

4. A Shared Access Signature (SAS) token was generated with Read permissions and a specific time-bound expiration.

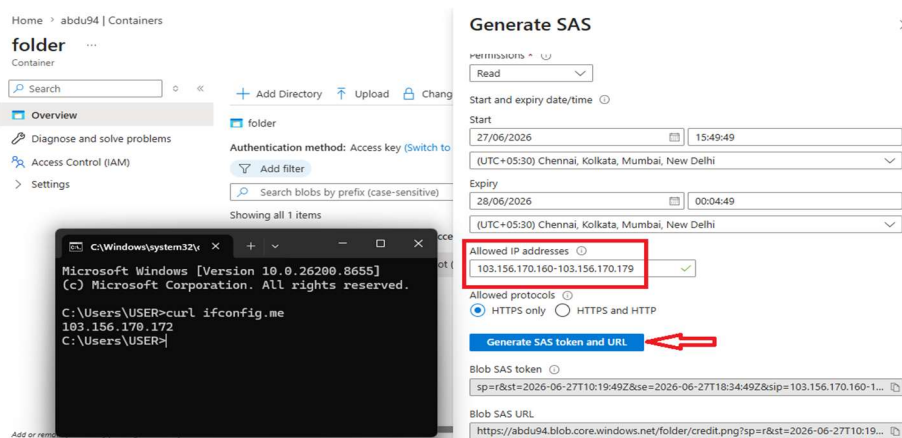


Figure 5: Generating blob SAS token specifying Read permission and custom start/end timestamps.

5. Access was validated using the SAS URL in a browser. The file successfully displayed, confirming authorized access.

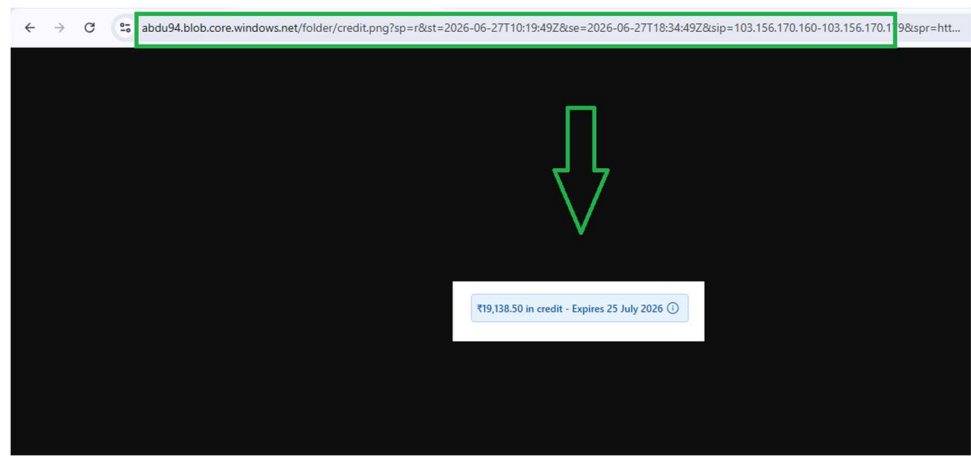


Figure 6: Successful rendering of file.pdf when accessed via the generated SAS token URL.

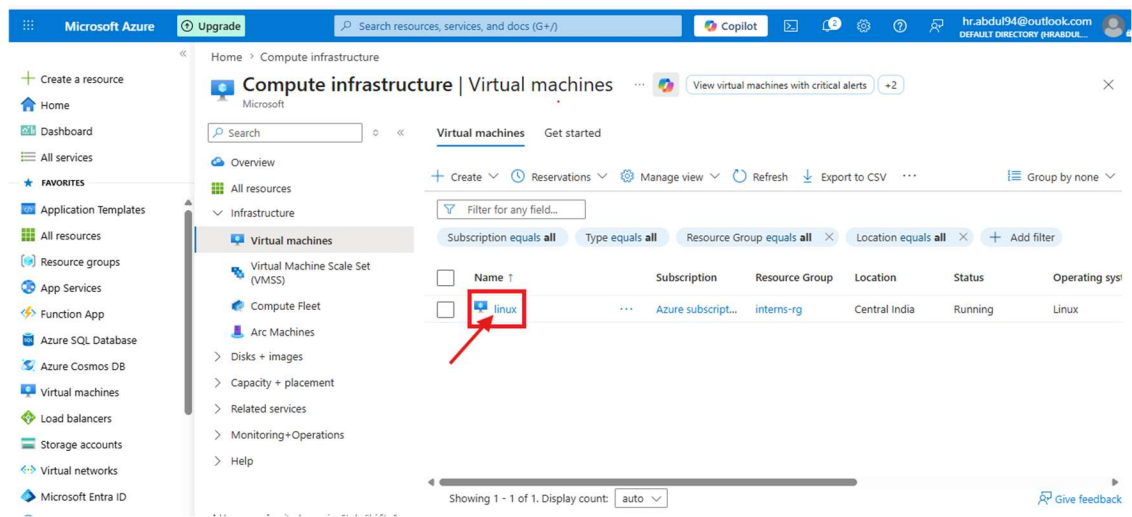
Task 2: Secure Virtual Machine Deployment & Hardening

A Linux-based Virtual Machine was provisioned. To reduce exposure to automated scanning tools, standard SSH access via port 22 was hardened by reconfiguring the server to use port 2244 and creating custom firewall rules.

Parameter	Configuration Detail
VM Name	linux
Resource Group	interns-rg
OS Image	SLES 15 SP5 (Initial) / Ubuntu 24.04 (Final)
IP Address	20.244.6.120
Hardened SSH Port	2244

Step-by-step Execution and Evidence:

1. Virtual Machine was provisioned successfully in the Portal. The state was checked and confirmed as Running.



2. System properties, IP specification, size details, and billing models were verified on the VM essentials tab.

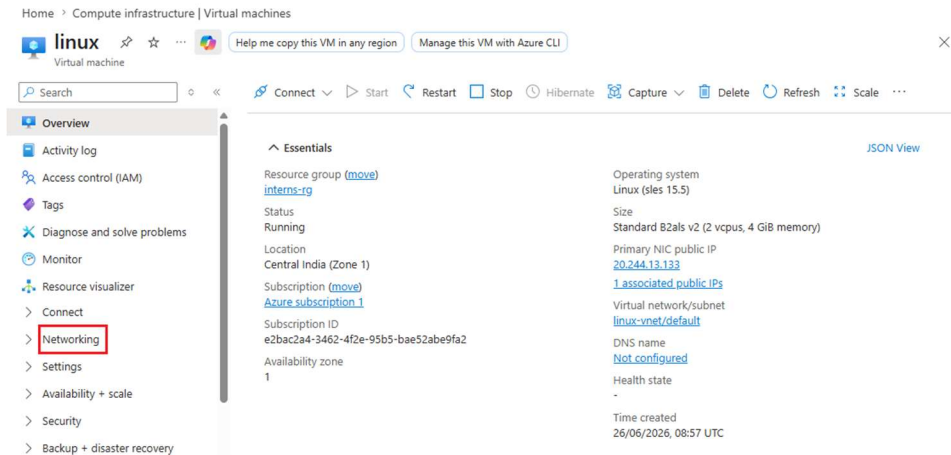


Figure 2: Essentials panel displaying details of the VM.

3. An initial baseline test confirmed port 22 was open and responsive to administrative connections.

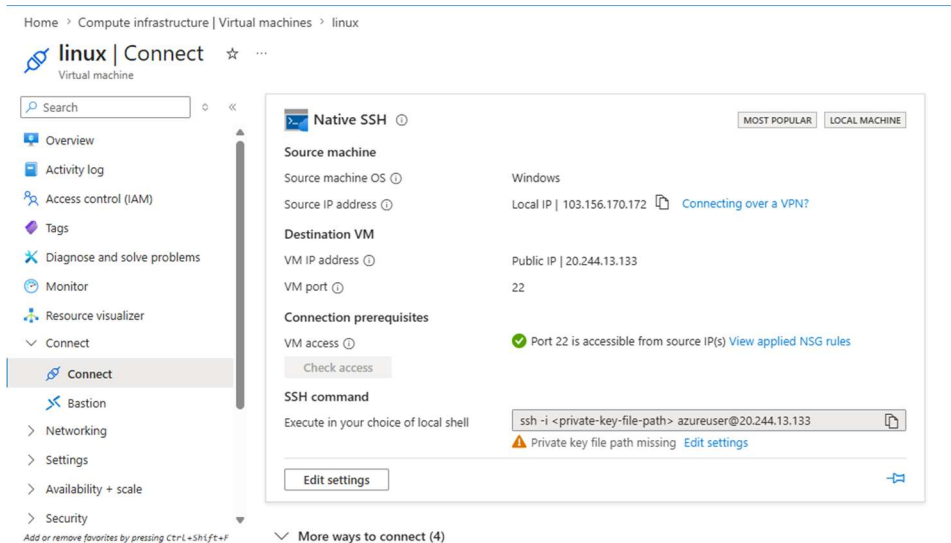


Figure 3: Connection diagnostic confirming standard port 22 is reachable.

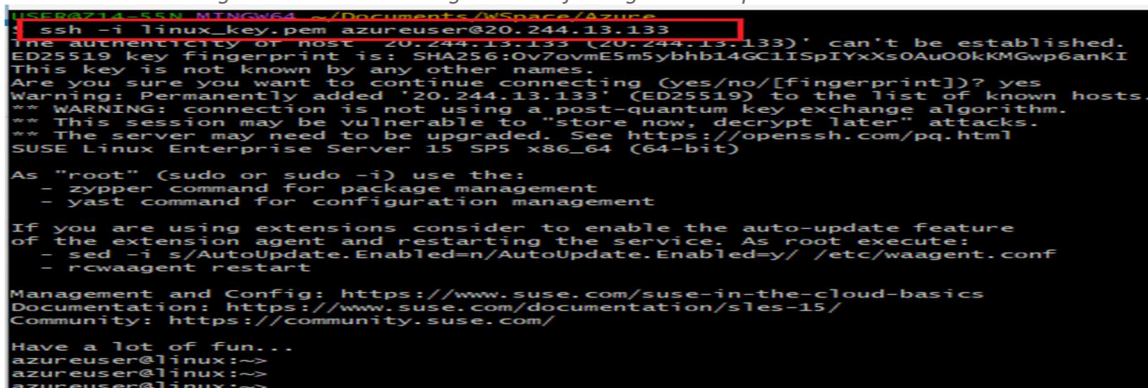


Figure 4: Terminal console displaying successful baseline login on default port 22.

4. The host server configurations were hardened. Port 22 was blocked, and a custom rule allowed traffic on port 2244.

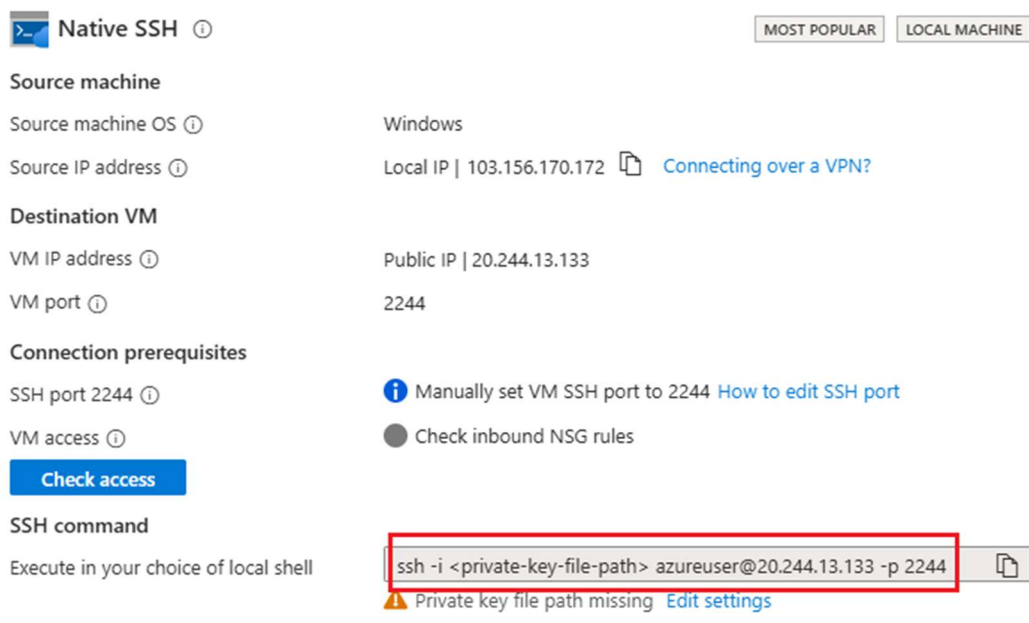


Figure 5: Azure Portal - Defining parameters to open Custom SSH Port 2244.

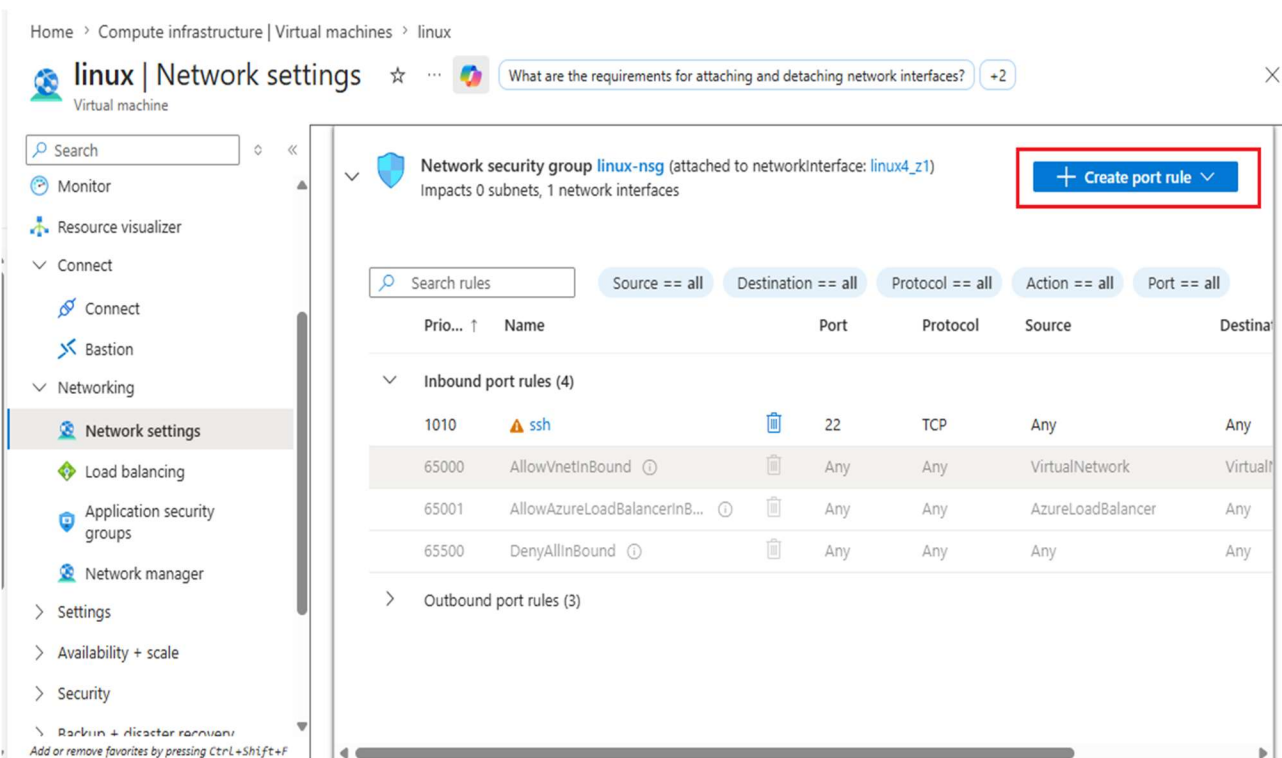


Figure 6: NSG inbound rule lists showing Rule active Port 22 default rules.

5. Hardening was verified. Standard port 22 connections were refused, and successful connections were logged on custom port 2244.

The screenshot shows the configuration for a Network Security Group (NSG) rule. At the top, a dropdown menu is set to 'Custom'. Below it, the 'Destination port ranges' field is highlighted with a red box and contains the value '2244', which is also marked with a green checkmark. The 'Protocol' section has radio buttons for 'Any', 'TCP' (which is selected), 'UDP', 'ICMPv4', and 'ICMPv6'. The 'Action' section has radio buttons for 'Allow' (which is selected) and 'Deny'. The 'Priority' field at the bottom is set to '1009' and is also marked with a green checkmark.

Figure 7: NSG configuration Port 2244 is only allowed .

```
USER@714-SSN-MTNGW64 ~ /Documents/Azure
$ ssh -i linux_key.pem azureuser@20.244.6.120 -p 2244
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1018-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Jun 27 06:15:59 UTC 2026

System load: 0.0          Processes:              122
Usage of /:  6.4% of 28.02GB Users logged in:             0
Memory usage: 9%          IPv4 address for eth0: 10.1.0.4
Swap usage:  0%

Expanded Security Maintenance for Applications is enabled.

0 updates can be applied immediately.

Last login: Sat Jun 27 05:59:03 2026 from 103.156.170.170
azureuser@linux:~$
```

Figure 8: Terminal SSH session successfully established on custom port 2244.

Section C: Security Analysis

Storage Security Analysis

1. Why was a private container used?

A private container acts as a primary barrier against data leakage. If containers are set to public, proprietary databases, keys, or logs are exposed to automated search bots indexing public S3 buckets and Azure blobs. Keeping containers private ensures that only authenticated corporate identities and signed client requests can touch the stored resources.

2. Why was SAS chosen instead of public access?

Shared Access Signature (SAS) tokens represent a granular, controlled approach to data sharing. SAS URLs allow clients or third-party developers to fetch data without exposing root access keys or giving them permanent Azure directory logins. The access is temporary, and the capability is strictly bound (e.g., Read-only, limiting write/delete risks). If a SAS token gets compromised, its lifespan is limited, and administrators can instantly block it by rotating the storage access keys.

Virtual Machine Security Analysis

1. Why are NSG rules important?

Network Security Groups (NSGs) act as virtual firewalls at the host network layer. They inspect and filter all inbound and outbound traffic. Without custom NSG rules, standard ports are exposed to the open internet, leaving virtual machines susceptible to continuous scanning and brute-force logins.

2. What risks are reduced by restricting inbound access?

By disabling default management ports (blocking port 22) and changing to custom ports (such as 2244) with specific allowed source IP rules, the following risks are mitigated:

- **Brute-force SSH dictionary attacks:** Automated malware bots scan the default port 22. Changing the port stops massive volumes of automated login attempts.
- **Service exploitation:** It reduces the risk of attackers exploiting zero-day security vulnerabilities in the SSH daemon.
- **Denial of Service (DoS):** Restricts open port network traffic to authorized IPs, preventing resource exhaustion from scanning spikes.

Recommendations & Future Actions

To build upon this baseline security environment, the following future implementations are recommended for XYZ Solutions:

- **Enable Just-in-Time (JIT) VM Access:** Implement JIT VM Access through Microsoft Defender for Cloud. JIT blocks administrative ports (like custom port 2244) by default. When admins need access, they request it. Defender opens the firewall only for their specific IP and closes it automatically after a limited timeframe (e.g., 3 hours).
- **Integrate Microsoft Sentinel and Log Collection:** Link the Log Analytics Workspace to Microsoft Sentinel. Configure syslog log sources and data collection rules to forward authentication logs. Create alert detection rules. For example, trigger alerts on brute-force attempts using this KQL

query:

```
Syslog
| where ProcessName == "sshd" and SyslogMessage has "Failed password"
| summarize FailedAttempts = count() by SourceIP = extract("from ([0-9.]+)", 1, SyslogMessage), Computer, TimeGenerated
| where FailedAttempts > 5
```

- **Enforce Multi-Factor Authentication (MFA):** Configure Microsoft Entra ID Conditional Access policies to require MFA authentication for 'infrauser', 'storageuser', and all cloud administrators, preventing compromise via credential leakages.
- **Restrict Storage Accounts via Private Endpoints:** Remove storage account public network paths entirely. Configure a Private Endpoint inside a secured Virtual Network (VNet) to ensure storage resources are only accessible within the private corporate network scope.